

Rajdeep Gill

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Education

University of Manitoba

Bachelor of Science in Computer Engineering

- Recipient of the University Medal for highest academic standing in Computer Engineering

June 2026

4.46/4.50 GPA

Skills

Programming Languages: TypeScript, JavaScript, Python, C++, Java

Frameworks and Libraries: React, NextJS, PyTorch, OpenMPI, Node.js, Express, Flask

Tools: Git, Docker, PostgreSQL, Github Actions, Bazel, gRPC, Protocol Buffers (Protobuf), Amazon Web Services (AWS), Google Cloud Platform (GCP)

Experience

Robinhood | *Software Engineer*

Sep 2025 - Current

- Led a team of 3 engineers to build a tax lot selling feature, implementing observables for real-time market data updates and virtualization to ensure performant rendering for users with thousands of positions, driving \$55M in notional volume
- Owned the margin web experience end-to-end as the team's sole frontend engineer, shipping a native onboarding flow that cut onboarding time 70% and lifted enrollment 12%, and redesigning the margin call interface with clear risk indicators and actionable recovery steps to reduce fund recovery time
- Built a highly composable set of React components for interactive data tables supporting virtualization, sorting, filtering, column pinning, and pagination, becoming the standard building block for future table-based features across the team
- Enhanced an internal cross-platform library to bridge React context and Redux stores into Web Workers, enabling shared state and cache access for parallelized processing without blocking the main thread

Robinhood | *Software Engineer Intern*

May 2025 - Sept 2025

- Owned end-to-end development and delivery of a real-time margin maintenance table built with React and TypeScript, featuring embedded trade execution and risk indicators
- Overhauled the margin experience by expanding the server-driven UI to include futures traders and unifying business logic across Web, iOS, and Android, driving a 43% increase in visitors to the margin experience
- Enhanced a Protobuf-based server-driven UI architecture by implementing 12 foundational components in Python and Typescript along with a logging framework to increase system usage and maintainability
- Repaired the core integration suite for the margin experience and added 20+ test suites, achieving 93% code coverage

Biosensors Research Lab | *Machine Learning Research Intern*

May 2024 - Dec 2024

- Created a custom convolutional neural network to predict cell viability from optical microscopy images, achieving 85% accuracy, establishing a non-invasive alternative to traditional staining methods for assessing cell health
- Developed a model validation GUI in Python displaying SHAP-based feature attributions, and visualizing cell images to debug model predictions
- Created a parallelized data processing pipeline with a new morphological parameter extraction algorithm, improving accuracy by 17% and cutting processing runtime by 75%.
- Replaced a high-risk system of manually-shared hard drives with a centralized MySQL database, allowing for cross-experiment queries and easy data sharing

Biosensors Research Lab | *Machine Learning Research Intern*

May 2023 - Aug 2023

- Built a framework for electronic testing equipment, allowing for automation of entire experimental sequences through Python scripts and cutting manual setup time
- Programmed an FPGA to coordinate microfluidic device operations, integrating di-electrophoresis control with synchronized LED illumination and cell imaging

Projects

Distributed Image Processing (C++, OpenMPI)

- Accelerated the processing of gigapixel-scale images by 3.8x on 4 nodes by designing a scalable OpenMPI pipeline allowing data to be partitioned across an N-node cluster

AI Bug Trace (Python, FastAPI, LangChain, Docker)

- Built a dockerized backend service to track and evaluate developer debugging workflows by capturing file operations, terminal commands, test results and pull requests to collect better data for fine-tuning LLM models